Correlated Country Risk and International Capital Allocation

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Abstract

International investors treat neighboring countries as a package: they favor or avoid them together. We show that this neighborhood structure is a first-order force in the global allocation of capital, distinct from the familiar cost of investing far from home. We develop a multi-country general-equilibrium model of international portfolio allocation that separates the two. We identify the neighborhood component from how attractiveness, revealed by portfolio holdings, co-moves across countries. It accounts for two-thirds of persistent cross-country differences in attractiveness. Sovereign credit risk and geopolitical alignment share its spatial structure, pointing to geographic patterns in risk and bloc formation as its sources. Geography shapes where capital ends up. Together, the two channels account for 64 percent of the cross-country variance of log capital and 25 percent of that of log output, more than the sum of their separate contributions. The neighborhood component also governs how shocks travel. When confidence in one country falls, capital flees to distant economies and lowers their interest rates, but rates in its neighbors rise. Geography links destinations to one another, not only investors to destinations.

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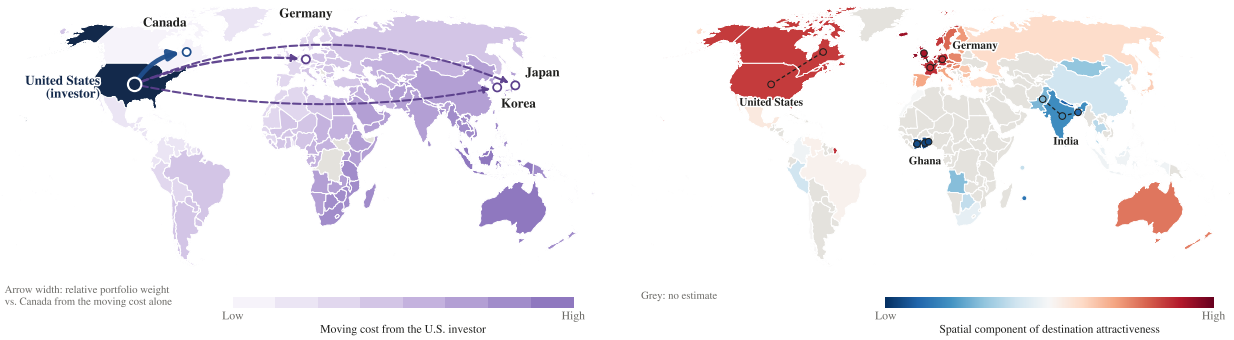
1 Introduction

Geographic distance continues to shape international investment despite substantial reductions in the technological cost of moving capital across borders. Investors disproportionately hold domestic and nearby foreign assets, and bilateral portfolio positions decline with distance. These patterns are commonly interpreted as evidence of information frictions, familiarity, or other barriers to international investment. Yet geography may matter for international capital allocation in another way. Countries located near one another may also share persistent characteristics that make them similar investment destinations. These two forms of geographic dependence are observationally related but economically distinct.

This paper studies how these two geographic forces shape international capital allocation and cross-country interest-rate differences. We begin by documenting that bilateral interest-rate differentials increase with geographic distance. The relationship remains when we isolate the persistent country-pair component of interest-rate differences after removing country-specific time-varying shocks. Thus, the geographic pattern in interest rates is not solely a consequence of transitory regional or macroeconomic fluctuations.

We distinguish two channels through which geography can generate this pattern. The first operates between an investor and a destination. Investors face bilateral portfolio frictions that increase with geographic distance, so that otherwise similar investment opportunities receive less capital when they are farther from the investor. The second operates across destinations themselves. Destination-specific characteristics that affect investment demand may be spatially correlated, so that geographically proximate countries exhibit similar attractiveness to investors from all origins. The distinction is economically important. For example, a U.S. investor may allocate less capital to Korea than to Canada because Korea is farther from the United States. This is a bilateral investor–destination friction. At the same time, Korea may share persistent destination characteristics with nearby countries such as Japan. This second form of geographic dependence does not depend on the location of any particular investor. Investors from different origins may view nearby destinations as similar because their underlying investment characteristics are spatially correlated.

To quantify these channels, we develop a multi-country production economy with endogenous international portfolio allocation. Households save through competitive financial intermediaries



(a) Bilateral Portfolio Frictions (b) Spatially Correlated Destination Attractiveness

Figure 1: Geography Matters Through Two Distinct Channels

that allocate wealth across domestic and foreign destinations. Portfolio demand depends on the destination return, a bilateral portfolio wedge, and a destination-specific attractiveness component common to investors from all origins. On the production side, capital has a decreasing marginal product, so international portfolio allocation and equilibrium interest rates are jointly determined. The model provides a direct link between destination attractiveness and equilibrium interest rates. An increase in a country's attractiveness raises investment demand and induces capital inflows, which increase its capital stock and lower its equilibrium interest rate. Consequently, when destination attractiveness contains a spatially correlated component, geographically proximate countries tend to exhibit more similar interest rate responses.

The model also highlights how the geographic structure of attractiveness shocks affects the international transmission of interest rates. A negative idiosyncratic attractiveness shock induces capital outflows from the affected country, raising its interest rate while lowering interest rates elsewhere through capital reallocation. In contrast, when the shock is spatially correlated, nearby destinations also experience declines in attractiveness. Their interest rates may therefore rise rather than fall. Thus, spatial dependence in destination attractiveness can change not only the magnitude of international interest rate responses but also their direction.

We estimate the bilateral portfolio wedge using international portfolio holdings. The portfolio-allocation equation generates a gravity relationship in which holdings decline with bilateral distance after controlling for investor-by-time and destination-by-time factors. We parameterize bilateral frictions using geographic distance, home bias, and geopolitical alignment, and find that portfolio holdings decline with geographic distance and are lower across geopolitical blocs.

Because equilibrium returns are jointly determined with portfolio allocation, we estimate the sensitivity of portfolio demand to returns using an instrumental-variable strategy that builds on Koijen and Yogo (2026). The instrument compares predicted demand pressure with predicted asset supply. Predicted demand is constructed from the estimated bilateral gravity structure using demand originating from investors other than the focal investor, while predicted supply is based on observable destination characteristics. The resulting variation in relative demand pressure shifts equilibrium returns through market clearing and identifies the return sensitivity of portfolio demand.

We next study the destination-side channel. We recover destination-specific attractiveness and first show that geographically proximate countries tend to have more similar destination components. This relationship remains after removing observable differences in market size and is not eliminated by controlling for geopolitical distance. We then decompose the residual destination component into a persistent spatial component, a persistent non-spatial country component, and transitory variation. The spatial covariance is modeled as a decreasing function of bilateral distance and estimated jointly with the remaining variance components by restricted maximum likelihood. The estimates indicate substantial persistent spatial dependence in destination attractiveness, with the spatial component exceeding the non-spatial component of persistent cross-country heterogeneity.

We use the estimated model to quantify the implications of these two geographic channels through three exercises. First, we hold investor-country asset supplies fixed and decompose population-weighted interest-rate dispersion relative to the United States. We find that spatially correlated destination attractiveness accounts for most of the aggregate geographic contribution to interest-rate dispersion, while the contribution of bilateral geographic frictions is relatively small. However, this aggregate result masks substantial heterogeneity across country groups. Geographic forces reduce interest-rate dispersion relative to the United States among advanced economies but increase it among emerging market economies, with the latter effect driven primarily by spatially correlated destination attractiveness. Second, we examine how the geographic structure of destination attractiveness shocks shapes international interest rate transmission. We compare negative idiosyncratic shocks with spatially correlated shocks that also affect nearby destinations. We find that an idiosyncratic decline in destination attractiveness lowers interest rates in foreign countries through capital reallocation, whereas a spatially correlated decline can raise interest rates in geographically proximate countries while lowering them in more distant countries. These results highlight how spatial dependence

can alter the direction of international interest rate transmission. Third, we allow investor asset supplies, capital stocks, and equilibrium returns to adjust endogenously in steady state. Removing either bilateral geographic frictions or the spatial component of destination attractiveness reduces cross-country dispersion in capital and output, while removing both channels leads to substantially larger reductions. These results indicate that both geographic channels play quantitatively important roles in shaping the international distribution of capital and output.

Taken together, our findings show that geography in international finance cannot be characterized solely by bilateral barriers to investment. Geographic distance also organizes the characteristics of investment destinations, generating persistent spatial dependence that affects equilibrium interest rates and the international transmission of financial shocks. The two geographic channels operate through distinct mechanisms and have different quantitative implications. Spatially correlated destination attractiveness accounts for most of the aggregate geographic contribution to interest rate dispersion and can reverse the direction of international interest rate transmission, while both channels substantially shape the long-run distribution of capital and output.

Our paper contributes to the literature in three ways. First, we connect the geographic structure of international portfolio holdings to persistent cross-country interest rate differences in a general-equilibrium model. Second, we distinguish bilateral investor–destination frictions from spatial dependence across destinations and estimate the latter directly from the covariance structure of destination attractiveness. Third, we quantify how these geographic channels affect interest rate dispersion, international interest rate transmission, and the allocation of capital and output, highlighting their distinct implications across outcomes and country groups.

The remainder of the paper proceeds as follows. Section 2 describes the data. Section 3 documents the geographic gradient in bilateral interest rate differentials and its persistent component. Section 4 develops the model and characterizes the equilibrium interest rate responses to destination attractiveness. Section 5 estimates bilateral portfolio frictions and the sensitivity of portfolio demand to returns. Section 6 estimates and recovers the spatial component of destination attractiveness. Section 7 presents the quantitative decomposition, international shock-transmission analysis, and steady state counterfactuals. Section 8 concludes.

Related Literatures Our paper relates to several strands of the literature. First, it contributes to the open-economy macroeconomics literature studying international capital allocation and global portfolio positions, including Lucas (1990), Obstfeld and Rogoff (1995), Gourinchas and Jeanne (2006), Pavlova and Rigobon (2007), Gourinchas and Rey (2007), Caselli and Feyrer (2007), Alfaro et al. (2008), Pellegrino et al. (2025), and Kleinman et al. (2026). Our analysis is also related to models of segmented international financial markets, including Gabaix and Maggiori (2015), Maggiori (2017), and Itskhoki and Mukhin (2021, 2025), which emphasize frictions that prevent capital from being allocated frictionlessly across countries.

Second, our analysis relates to the literature documenting gravity-type patterns in international capital allocation. Portes et al. (2001) and Portes and Rey (2005) show that international financial transactions decline with geographic distance, with related evidence provided by Aviat and Coeurdacier (2007) and Lane and Milesi-Ferretti (2008). Coeurdacier and Martin (2009) and Okawa and van Wincoop (2012) incorporate gravity-type portfolio relationships into models of international asset allocation. More recently, Koijen and Yogo (2026) exploit gravity-type variation in asset demand to construct instruments for estimating the sensitivity of international portfolio demand to returns. Our paper builds on this literature by treating geographic distance not only as a bilateral portfolio friction but also as a source of spatial structure in destination-side investment attractiveness.

Third, the paper relates to work studying spatial dependence in macroeconomics and finance, including LeSage and Llano (2013), Dell'Erba et al. (2013), Lustig and Richmond (2020), Servén and Abate (2020), Krisztin and Piribauer (2021), and Shambaugh and Zhou (2024). This literature highlights that macroeconomic and financial outcomes can exhibit systematic dependence across geographically connected economies. Our contribution is to bring this spatial perspective into a model of international portfolio allocation by allowing destination-specific attractiveness to contain a spatially correlated persistent component.

In terms of modeling, our framework is related to dynamic spatial equilibrium models, including Allen and Arkolakis (2025) and Eckert and Kleineberg (2024), as well as the asset demand system developed by Koijen and Yogo (2019, 2026). More specifically, our model is closely related to Kleinman et al. (2026) and Pellegrino et al. (2025), who embed international asset demand into open-economy general equilibrium models to study the allocation of capital across countries. Pellegrino et al. (2025), in particular, study how bilateral geographic frictions, taxation, and political risk

shape global capital allocation. Our paper instead studies how geography shapes international capital allocation through two distinct channels: bilateral investor–destination portfolio frictions and spatially correlated destination-specific heterogeneity. We then quantify the contribution of each channel to cross-country interest rate dispersion.

2 Data

Interest Rates and Inflation Interest-rate data are obtained from the Global Macro Database. To measure returns on capital while minimizing the influence of short-run monetary policy movements, we use long-term interest rates. For countries with incomplete coverage in the Global Macro Database, including China, Indonesia, Peru, and the Philippines, we supplement the series with 10-year government bond yields from Covered Interest Rate Parity Deviations Between Government Bonds by Wenxin Du. Real interest rates are constructed using five-year-ahead inflation expectations from the IMF World Economic Outlook Historical Forecasts.

Bilateral Capital Holdings We construct bilateral portfolio holdings as the sum of short- and long-term debt securities and equity securities. Bilateral holdings are obtained from the IMF Coordinated Portfolio Investment Survey (CPIS). To construct total amounts outstanding, we combine information from the OECD, BIS Debt Securities Statistics, and World Bank market capitalization data. For the United States, we supplement the CPIS data with Treasury International Capital (TIC) data. To address distortions arising from financial centers and tax havens, we reallocate portfolio positions from a residency basis to an ultimate nationality basis using the Global Capital Allocation Project of Coppola et al. (2021). The implementation follows the data construction in the replication package of Koijen and Yogo (2026).

Country Characteristics We compile country-level and bilateral characteristics from several sources. Sovereign credit default swap (CDS) spreads and sovereign credit rating scores are obtained from Refinitiv Datastream. Capital-account openness is measured using the Chinn–Ito index, and political stability is obtained from the World Bank Worldwide Governance Indicators. To capture geopolitical alignment, we use United Nations General Assembly ideal-point estimates from Bailey, Strezhnev, and Voeten and geopolitical bloc classifications from the replication package of Gopinath

et al. (2025). Bilateral geographic distance and other country-pair characteristics are obtained from CEPII.

3 Motivating Empirical Evidence

This section investigates the spatial relationship in interest rates across countries. If frictions to international capital allocation have a geographic dimension, or if factors relevant for interest rates are spatially correlated, interest rates should exhibit a systematic geographic pattern across countries. In particular, countries that are geographically closer may display more similar interest rates, whereas interest-rate differentials may increase with bilateral distance. To examine this relationship, we consider two empirical specifications. The first exploits time variation in bilateral interest-rate differentials, while the second isolates their time-invariant, persistent component.

3.1 Geographic Gradient in Interest-Rate Differentials

We consider following regression.

$$|r_{i,t} - r_{j,t}| = \beta \ln(dist_{i,j}) + X_{i,j}\Gamma + \alpha_i + \alpha_j + \lambda_t + \varepsilon_{ijt}$$

For each country pair, the interest-rate differential is defined as absolute value of rate differences. Interest rates are measured in percentage points, so that a one-unit difference in the dependent variable corresponds to a one-percentage-point interest-rate gap. The vector $X_{i,j}$ contains bilateral controls, including indicators for whether the two countries share a border or a common language, have a colonial relationship, or were historically part of the same country. The specification also includes fixed effects for the origin- and destination countries, α_i and α_j, respectively, as well as year fixed effects λ_t. The coefficient of interest, β, measures how the interest-rate gap varies with geographic distance. A positive value of β indicates that the interest-rate differential tends to widen as the geographic distance between the two countries increases. The high- and low-country fixed effects absorb systematic differences associated with each country depending on its position in the interest-rate ranking. Intuitively, the country fixed effects allow us to compare how the interest-rate gap varies across partners of a given country. For example, conditional on the fixed effect associated with the United States, the regression compares whether the U.S. interest-rate gap tends to be larger

with geographically more distant countries.

Panel A in Table 1 shows that geographically more distant country pairs tend to exhibit larger interest rate differentials. The coefficient on geographic distance remains positive and statistically significant after controlling for bilateral characteristics, including shared borders, common language, colonial ties, and historical state relationships. These findings indicate a robust geographic gradient in bilateral interest rate differences.

3.2 The Persistent Component of Interest-Rate Differentials

The baseline specification establishes that interest-rate differentials are larger between geographically distant countries. However, this relationship may partly reflect transitory country-specific shocks. For example, geographically proximate countries may experience similar monetary or macroeconomic shocks in a given year, generating temporarily similar interest rates. We therefore examine whether geographic distance is also related to the persistent component of bilateral interest-rate differentials. We first decompose the bilateral interest-rate differential into origin-country-by-year and destination-country-by-year components and a time-invariant country-pair component, $\delta_{i,j}$. The country-by-year fixed effects absorb time-varying shocks specific to either member of the pair. The estimated pair component therefore captures the portion of the bilateral interest-rate differential that persists over time after removing these country-specific fluctuations.

$$|r_{i,t} - r_{j,t}| = \alpha_{i,t} + \alpha_{j,t} + \underbrace{\delta_{i,j}}_{\text{pair FE}} + u_{i,j,t}$$

In the second stage, I regress the estimated pair component on bilateral geographic distance and the same set of bilateral controls used in the baseline specification.

$$\hat{\delta}_{i,j} = \beta^P \ln(dist_{i,j}) + X_{i,j}\Gamma + \eta_{i,j}$$

Panel B in Table 1 reports the regression estimates, while Figure 2 (b) presents the corresponding binscatter of the persistent interest-rate differential against log bilateral distance. Both show a strong positive geographic gradient. The baseline estimate of β^P implies that bilateral distance is associated with a larger persistent interest-rate gap. The coefficient remains similar across specifications that control for bilateral relationships.

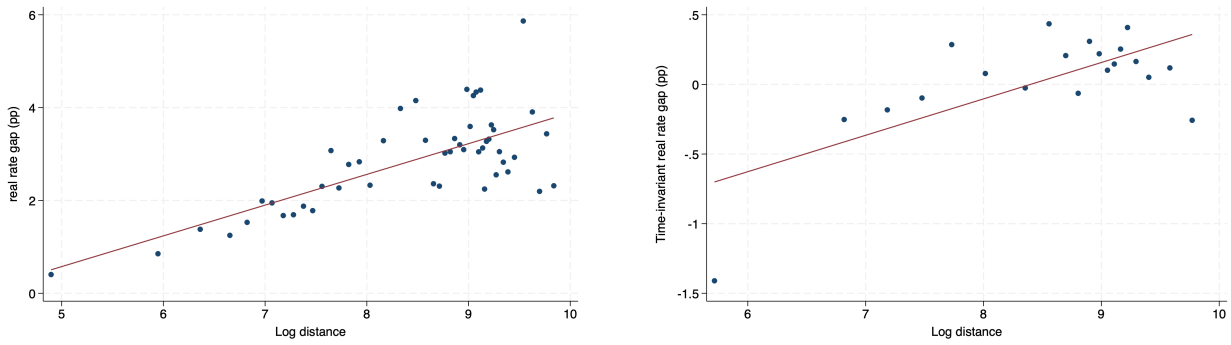
Table 1: Geographic Distance and Interest-Rate Differentials

Panel A. Pooled Panel

	(1)	(2)	(3)	(4)
Log distance	0.232***	0.228***	0.242***	0.236***
	(0.039)	(0.047)	(0.036)	(0.045)
Shares border		-0.036		-0.109
		(0.120)		(0.128)
Common language			0.104	0.076
			(0.096)	(0.105)
Colonial relationship				0.208*
				(0.109)
Same country (past/now)				0.121
				(0.160)
Origin FE	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Observations	56088	56088	56088	56088
Adj. R-squared	0.504	0.504	0.504	0.504

Panel B. Persistent Pair Component

	(1)	(2)	(3)	(4)
Log distance	0.261***	0.275***	0.261***	0.283***
	(0.023)	(0.025)	(0.023)	(0.025)
Shares border		0.226		0.101
		(0.146)		(0.155)
Common language			-0.004	-0.082
			(0.079)	(0.082)
Colonial relationship				0.447***
				(0.158)
Same country (past/now)				0.354
				(0.244)
Observations	4323	4323	4323	4323
Adj. R-squared	0.028	0.029	0.028	0.030



(a) Pooled panel (b) Persistent component

Figure 2: Interest-Rate Differentials and Geographic Distance

4 Model

In this section, we develop a dynamic spatial production economy with multiple countries. There are N countries, indexed by $n \in \{1, \ldots, N\}$. Each country is populated by a representative household and competitive firms. Households choose consumption and saving, while their savings are allocated across domestic and foreign investment opportunities through financial intermediaries. International portfolio allocation is subject to bilateral frictions, so that otherwise identical investment opportunities may be valued differently across investor–destination pairs. This setup is consistent with the recent literature on segmented international capital markets, following Gabaix and Maggiori (2015) and Itskhoki and Mukhin (2021, 2025), and is closely related to the international portfolio-allocation frameworks in Pellegrino et al. (2025) and Kleinman et.al. (2026).

4.1 Households

The representative household in country n is endowed with a fixed amount of labor, ℓ_n. The household chooses consumption and saving to maximize lifetime utility,

$$\max_{\{c_{nt}, a_{n,t+1}\}} \sum_{t=0}^{\infty} \beta_n^t \, u(c_{nt}) \tag{1}$$

$$\text{subject to } c_{n,t} + a_{n,t+1} = \mathcal{R}_{n,t} a_{n,t} + w_{n,t} \ell_n$$

where β_n is the country-specific discount factor,[1] and $\mathcal{R}_{n,t}$ denotes the return on the household's savings portfolio. The optimal consumption-saving decision satisfies $u'(c_{n,t}) = \beta_n \mathcal{R}_{n,t+1} u'(c_{n,t+1})$. In steady state, consumption and portfolio returns are constant over time, so the Euler equation reduces to $1 = \beta_n \mathcal{R}_n$.

4.2 International Capital Allocation

An investor in country n considering investment in country i faces a bilateral portfolio wedge Φ_{ni}. The wedge captures frictions that reduce the relative attractiveness of investing in a particular foreign destination. We allow the wedge to depend on both geographic distance and geopolitical alignment. First, we assume that the wedge increases with bilateral geographic distance, reflecting informational

[1] Cross-country differences in discount factors may reflect differences in mortality risk, as in Pellegrino et al. (2025), or differences in time preferences, as documented by Falk et al. (2018).

frictions, limited familiarity, or other barriers associated with investing in more distant markets. Second, motivated by the recent literature on geoeconomic fragmentation, which documents that cross-border FDI and portfolio holdings tend to be lower between geopolitically misaligned countries (e.g., Gopinath et al., 2025), we allow the wedge to increase when the investor and destination belong to different geopolitical blocs. Thus, Φ_{ni} captures both geographic and geopolitical barriers to international capital allocation.

The term ω_{it} captures destination-specific attractiveness that is common to investors from all origin countries. Unlike the bilateral portfolio wedge Φ_{ni}, which varies across investor–destination pairs, ω_{it} depends only on the destination country i and time t. It therefore captures characteristics of country i that affect its attractiveness as an investment destination uniformly across investors, beyond the observed interest rate r_{it}. A higher ω_{it} increases the portfolio share allocated to country i by investors from all countries. One potential source of destination attractiveness is country credit risk. Countries with lower credit risk may be more attractive to international investors, holding observed returns fixed. This distinction is important for the analysis below. While Φ_{ni} captures bilateral segmentation between investor n and destination i, ω_{it} captures destination-side heterogeneity that is common across investors.

For each unit of savings, the intermediary draws an idiosyncratic taste shock ε_{ni} for each destination i, independently across destinations, from a Gumbel distribution.

$$F(x) = \exp(-\exp(-x + \bar{\gamma}))$$

where $\bar{\gamma}$ is Euler–Mascheroni constant. The scale of the idiosyncratic shock is governed by σ_N. The intermediary in country n invests in country i if the friction- and preference-adjusted return from investing in i is the highest among all available destinations.

$$\varepsilon(n,i,\nu)\sigma_N - \Phi_{ni} + \omega_{it} + r_{i,t} = \max_h \{\varepsilon(n,i,\nu)\sigma_N - \Phi_{nh} + \omega_{ht} + r_{h,t}\} \tag{2}$$

The share of country n's savings allocated to country i through financial intermediaries is given by

$$b_{nit} = \frac{a_{nit}}{a_{nt}} = \frac{\exp(-\Phi_{ni} + \omega_i + r_{i,t})^{1/\sigma_N}}{\sum_h \exp(-\Phi_{nh} + \omega_h + r_{h,t})^{1/\sigma_N}} \tag{3}$$

Here, a_{nit} denotes the amount of country n's savings allocated to country i, while a_{nt} denotes total savings in country n. A larger portfolio wedge Φ_{ni} reduces the share allocated to destination i,

whereas a higher destination-specific attractiveness ω_{it} or a higher interest rate r_{it} increases its portfolio share. The parameter σ_N governs the dispersion of idiosyncratic investment preferences and, therefore, the sensitivity of portfolio allocation to differences in returns and destination characteristics. The gross return on the household's savings portfolio is then the portfolio-share-weighted average of destination returns:

$$\mathcal{R}_{n,t} = 1 + \sum_i b_{nit} r_{i,t} \tag{4}$$

4.3 Firm

Each country n is populated by a representative firm operating a Cobb–Douglas production technology. The firm takes the wage $w_{n,t}$ and the interest rate $r_{n,t}$ as given and chooses its capital stock to maximize the present value of profits. Output is produced using capital $k_{n,t}$ and labor $\ell_{n,t}$, with country-specific productivity $z_{n,t}$.

$$r_{n,t+1} = \alpha z_{n,t} k_{n,t}^{\alpha-1} \ell_{n,t}^{1-\alpha}$$

Thus, the interest rate in each country is decreasing in the amount of capital allocated to that country. An increase in capital inflows raises $k_{n,t}$, lowers the marginal product of capital, and thereby reduces the equilibrium interest rate. This relationship provides the link between international portfolio allocation and cross-country interest-rate differences in the model.

4.4 Market Equilibrium

A competitive equilibrium consists of allocations $\{a_{nt}, b_{nit}, k_{nt}\}_{n,i,t}$ and prices $\{\mathcal{R}_{nt}, r_{nt}\}_{n,t}$ such that households optimize given prices and, for all n and t,

$$r_{nt} = \alpha z_n k_{nt}^{\alpha-1}, \tag{5}$$

$$k_{nt} = \sum_i b_{int} a_{it}, \tag{6}$$

$$\mathcal{R}_{nt} = 1 + \sum_i b_{nit} r_{it}. \tag{7}$$

4.5 Destination Attractiveness and Equilibrium Interest Rates

We examine how destination attractiveness affects equilibrium interest rates through international capital allocation. Consider an equilibrium in which investor asset stocks $\{A_n\}$ are fixed, while

portfolio shares, destination capital stocks, and interest rates adjust endogenously.

Proposition 1 (Destination Attractiveness and Interest Rates)**.** Consider an interior equilibrium with strictly positive portfolio shares and decreasing marginal products of capital. The first-order equilibrium response of interest rates to changes in destination attractiveness is given by

$$d\mathbf{r} = -P\, d\boldsymbol{\omega}, \qquad P = (H^{-1} + J)^{-1} J, \tag{8}$$

where P is the equilibrium interest-rate transmission matrix, and H and J characterize the responsiveness of capital supply and portfolio demand to interest rates, respectively.

The proof is provided in Appendix I.1.

An increase in destination attractiveness raises investment demand, inducing capital inflows that increase the destination's capital stock and lower its marginal product of capital. The resulting decline in the equilibrium interest rate partially offsets the initial increase in attractiveness. In the general multi-country economy, interest-rate responses depend on the entire portfolio allocation structure. This relationship provides a link between the spatial structure of destination attractiveness and geographic patterns in interest rates. If destination attractiveness is spatially correlated, geographically proximate countries tend to experience more similar attractiveness changes and, consequently, more similar equilibrium interest-rate responses.

We next examine how the geographic structure of destination attractiveness shocks affects the direction and magnitude of equilibrium interest-rate responses across countries. Using the equilibrium transmission matrix P derived above, we compare two types of attractiveness shocks originating in the same country i.

Idiosyncratic Shock. Consider a positive attractiveness shock of magnitude $h > 0$ that affects only country i. The shock and its equilibrium interest rate response in country j are given by

$$d\boldsymbol{\omega}_i^I = he_i, \qquad \frac{\partial r_j^I}{\partial h} = -P_{ji}, \tag{9}$$

where e_i is the i-th unit vector.

Under the equilibrium conditions, $0 < P_{ii} < 1$ and $P_{ji} < 0$ for $j \neq i$. Consequently, a positive idiosyncratic attractiveness shock lowers the interest rate in the origin country while raising interest

rates in other countries through international capital reallocation.

Spatially Correlated Shock. We next consider a shock originating in country i that also affects geographically proximate destinations. Let K denote a geographic kernel satisfying $K_{ii} = 1$, with K_{ji} declining in the distance between countries i and j. The spatial shock and its equilibrium interest rate response are given by

$$d\boldsymbol{\omega}_i^S = hKe_i, \qquad \frac{\partial \mathbf{r}^S}{\partial h} = -PKe_i. \tag{10}$$

For a recipient country $j \neq i$,

$$\frac{\partial r_j^S}{\partial h} = \underbrace{-P_{ji}}_{\text{Origin spillover}} - \underbrace{P_{jj}K_{ji}}_{\text{Own attractiveness}} - \underbrace{\sum_{k \neq i,j} P_{jk}K_{ki}}_{\text{Other-country spillovers}}. \tag{11}$$

Unlike an idiosyncratic shock, a spatially correlated shock can generate either positive or negative interest-rate responses in neighboring countries. In particular, the interest rate in country j declines if

$$P_{jj}K_{ji} > -P_{ji} - \sum_{k \neq i,j} P_{jk}K_{ki}. \tag{12}$$

Thus, the geographic structure of attractiveness shocks can affect not only the magnitude of international interest-rate responses but also their direction.

5 Parameterization

5.1 Portfolio Friction Estimation

This section estimates bilateral portfolio frictions, $\phi_{ni} = \Phi_{ni}/\sigma_N$, using PPML. From the portfolio allocation equation (3), investment shares can be decomposed into three components. The first is the bilateral portfolio friction, ϕ_{ni}, which varies across investor–destination pairs and captures barriers specific to international capital allocation. We parameterize this component as a function of bilateral geographic distance, home bias, and geopolitical alignment:

$$\phi_{ni} = \beta^{dist} \ln(dist_{ni}) + \beta^{foreign} \mathbb{1}[n \neq i] + \beta^{crossblock} Crossblock$$

The coefficient β^{dist} captures the geographic component of bilateral portfolio frictions. The coefficient $\beta^{foreign}$ captures home bias, or the additional friction associated with investing abroad relative to investing domestically. Finally, $\beta^{crossblock}$ captures the additional friction associated with investing in a destination that belongs to a different geopolitical bloc. Countries are classified into three groups—the U.S. bloc, the China bloc, and non-aligned countries—based on their ideal points in the United Nations General Assembly.

Second, the destination-specific component, $(\omega_{it}+r_{it})/\sigma_N$, is common across investors and is absorbed by destination-country $\times$ time fixed effects. Third, the denominator of the allocation equation captures investor-specific aggregate demand for foreign assets and is absorbed by investor-country $\times$ time fixed effects.

$$b_{nit} = \underbrace{\frac{\overbrace{\exp(-\phi_{ni})}^{\text{Capital Moving Friction}} \overbrace{\exp(\omega_{it}/\sigma_N + r_{it}/\sigma_N)}^{\text{Time}\times\text{Destin FE}}}{\sum_h \exp(-\phi_{nh} + \omega_{ht}/\sigma_N + r_{ht}/\sigma_N))}}_{\text{Time}\times\text{Origin FE}} \tag{13}$$

Estimation Results. The estimated coefficient on log geographic distance is -0.797, implying that doubling bilateral distance is associated with approximately 42 percent lower predicted portfolio holdings. The coefficient on the foreign indicator is -4.159, indicating substantial home bias in international capital allocation. The cross-bloc coefficient is -0.796, implying that portfolio holdings between countries belonging to different geopolitical blocs are approximately 55 percent lower, conditional on geographic distance and home bias.

These results suggest that both geographic distance and geopolitical alignment represent important bilateral frictions to international capital allocation. Appendix A.1 reports the detailed estimation results and robustness checks.

5.2 Estimation of Portfolio Demand Sensitivity

The sensitivity of portfolio demand to returns is governed by σ_N. Because equilibrium interest rates are jointly determined with international portfolio allocation, estimating this parameter requires an instrumental-variable strategy.

Following Koijen and Yogo (2026), we construct an instrument based on predicted asset demand

relative to predicted asset supply. Specifically, we use the estimated bilateral gravity structure to predict investment demand from all investor countries other than the focal investor. We then compare this predicted demand with asset supply predicted from observable destination characteristics. The resulting measure of residual demand pressure provides variation in equilibrium returns through asset-market clearing.

Using this instrument, we estimate the sensitivity of bilateral portfolio allocation to interest-rate differentials by two-stage least squares. The estimated return coefficient is 13.58, implying $\hat{\sigma}_N = 0.074$. Appendix A.2 provides the details of the instrument construction, identification assumptions, and estimation results.

5.3 Steady-State Calibration

The remaining country-specific parameters are calibrated to match the baseline steady state. We choose β_n to satisfy the steady-state Euler equation and calibrate z_n to match the observed interest rate. These parameters are held fixed in the counterfactual exercises.

6 Spatial Structure and Origins of Destination Attractiveness

6.1 Geographic Patterns in Destination Attractiveness

In the model, ω_{it} captures destination-specific attractiveness that is common to investors from all origin countries. If destination characteristics are spatially correlated, geographically proximate countries should exhibit more similar values of ω_{it}, generating geographic clusters in asset attractiveness. This provides a second channel through which geography affects international capital allocation, distinct from the bilateral portfolio friction ϕ_{ni}. The distinction is that ϕ_{ni} operates between an investor and a destination, whereas the spatial component of ω_{it} operates across destination countries themselves. For example, a U.S. investor may allocate less capital to Korea than to Canada because Korea is geographically more distant from the United States. This bilateral channel, however, does not capture the fact that Korea may share destination characteristics with nearby countries such as Japan. If ω_{it} is spatially correlated, investors from all origins may perceive Korean assets as more similar to Japanese assets than to assets in geographically distant countries such as Argentina. The spatial structure of ω_{it} captures this destination-side geographic dependence. To examine whether

destination attractiveness exhibits a geographic pattern, we first relate bilateral differences in ω_{it} to geographic distance:

$$|\omega_{it} - \omega_{jt}| = \beta^{\omega} \ln(dist_{ij}) + X'_{ij}\Gamma + \alpha_i + \alpha_j + \lambda_t + \varepsilon_{ijt}.$$

A positive coefficient on distance indicates that geographically distant countries tend to have more dissimilar destination components. Table 2a shows a positive and statistically significant relationship between bilateral distance and differences in ω_{it}. The relationship is robust to controls for contiguity, common language, colonial ties, and historical state relationships. Part of this relationship may reflect observable destination characteristics that themselves vary systematically across space. In particular, the recovered destination component may contain variation associated with market size. We therefore remove observable destination characteristics and year effects from ω_{it},

$$u_{it} = \omega_{it} - X'_{it}\hat{\Gamma} - \hat{\lambda}_t, \tag{14}$$

where the baseline specification includes log population in X_{it}. We then estimate the analogous bilateral regression

$$|u_{it} - u_{jt}| = \beta^{u} \ln(dist_{ij}) + Z'_{ij}\Pi + \alpha_i + \alpha_j + \lambda_t + \nu_{ijt}.$$

The positive relationship between geographic distance and differences in residual destination attractiveness remains after removing observable market size differences. Geopolitical distance is also significantly associated with differences in destination attractiveness, and controlling for it substantially attenuates the estimated geographic gradient. Nevertheless, geographic distance remains positive and statistically significant, suggesting that both geographic and geopolitical proximity are relevant to the spatial structure of destination attractiveness.

6.2 Spatial Decomposition

To characterize the spatial structure of destination attractiveness, we decompose the residual component u_{it} into a persistent spatial component, s_i, a persistent non-spatial component, a_i, and a transitory component, ε_{it}:

$$u_{it} = \underbrace{s_i}_{\substack{\text{Spatial} \\ \text{Component}}} + \underbrace{a_i}_{\substack{\text{Non-spatial} \\ \text{Component}}} + \varepsilon_{it}.$$

(a) Distance Regression on ω differences

	(1)	(2)	(3)	(4)
ln(distance)	0.0232***	0.0252***	0.0245***	0.0262***
	(0.0037)	(0.0041)	(0.0038)	(0.0042)
Contiguity		0.0237***		0.0138
		(0.0090)		(0.0090)
Common language			0.0263***	0.0238***
			(0.0072)	(0.0072)
Colonial link				0.0010
				(0.0105)
Same country				0.0193*
				(0.0102)
Country 1 FE	Yes	Yes	Yes	Yes
Country 2 FE	Yes	Yes	Yes	Yes
Year FE	YES	YES	YES	YES
Observations	57,100	57,100	57,100	57,100
Country pairs	4,324	4,324	4,324	4,324
Adj. R-squared	0.580	0.581	0.581	0.582

(b) Distance Regression on u differences

	(1)	(2)	(3)
ln(distance)	0.0263***		0.0099***
	(0.0015)		(0.0016)
Ideal-point distance		0.0620***	0.0551***
		(0.0028)	(0.0031)
Country 1 FE	Yes	Yes	Yes
Country 2 FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	57,100	57,100	57,100
Country pairs	4,324	4,324	4,324
Adj. R-squared	0.042	0.099	0.103

Table 2: Spatial relationships of destination attractiveness

As in a random-effects framework, we assume that the components are Gaussian

$$s \sim N(0, \Sigma_s), \qquad a \sim N(0, \sigma_a^2 I), \qquad \varepsilon_{it} \sim N(0, \sigma_\varepsilon^2).$$

To capture spatial dependence, we parameterize the spatial covariance matrix using a Gaussian covariance function [2]

$$[\Sigma_s]_{ij} = \sigma_s^2 \exp\left(-\left(\frac{d_{ij}}{\rho}\right)^2\right)$$

, where σ_s is scale parameter of spatial components while ρ is decay parameter which measures how quickly spatial dependence decay over distance.

6.3 Identification and Estimation of Parameters

Let $u_{it} = s_i + a_i + \varepsilon_{it}$ denote the latent deviation from the conditional mean. We assume that the spatial component, the non-spatial persistent component, and the transitory component are mutually orthogonal. We further assume that a_i is uncorrelated across countries and that ε_{it} is uncorrelated across countries and over time. Averaging over time gives

$$\bar{u}_i = \frac{1}{T_i} \sum_t u_{it} = s_i + a_i + \bar{\varepsilon}_i.$$

[2]The Gaussian covariance function, also known as the squared-exponential covariance function, can be obtained as the limiting case of the Matérn covariance family as the smoothness parameter tends to infinity.

The identifying information comes from covariance across countries and over time. For two different countries $i \neq j$,

$$\mathrm{Cov}(\bar{u}_i, \bar{u}_j) = \sigma_s^2 \exp\left[-\left(\frac{d_{ij}}{\rho}\right)^2\right],$$

where d_{ij} denotes chordal distance.[3] Under the orthogonality assumptions, cross-country covariance reflects only the spatial component. Its magnitude and variation with bilateral distance provide identifying information about σ_s^2 and ρ. Separate identification requires sufficiently informative variation in distances; identification can be weak when the spatial kernel is close to an identity matrix or a constant matrix.

For the same country observed in two different periods,

$$\mathrm{Cov}(u_{it}, u_{i\tau}) = \sigma_s^2 + \sigma_a^2, \qquad t \neq \tau.$$

Given the spatial variance, within-country persistence identifies σ_a^2. Finally,

$$\mathrm{Var}(u_{it}) = \sigma_s^2 + \sigma_a^2 + \sigma_\varepsilon^2.$$

The remaining variance after accounting for the two persistent components identifies σ_ε^2.

We jointly estimate $(\sigma_s^2, \rho, \sigma_a^2, \sigma_\varepsilon^2)$ by restricted maximum likelihood (REML), controlling for log population and year fixed effects. The estimation uses 1,028 country-year observations from 67 countries over 2003–2020. Table 3 reports the estimates. The estimated spatial variance is $\hat{\sigma}_s^2 = 0.01687$, while the non-spatial persistent and transitory variances are $\hat{\sigma}_a^2 = 0.00816$ and $\hat{\sigma}_\varepsilon^2 = 0.00352$, respectively.

The estimated spatial correlation range is $\hat{\rho} = 2{,}929$ km in chordal distance. Under the Gaussian covariance specification, correlation within the spatial component falls to 0.5 at a great-circle distance of approximately 2,454 km and to 0.1 at approximately 4,541 km.

A parametric-bootstrap likelihood-ratio test with 500 replications rejects the null hypothesis $H_0 : \sigma_s^2 = 0$, with an LR statistic of 36.609 and a bootstrap p-value of 0.002. This provides evidence against the absence of a spatial persistent component.

[3] Chordal distance is the Euclidean distance between two points on the Earth's surface embedded in three-dimensional space. Using this distance ensures that the Gaussian covariance kernel is positive semidefinite.

Table 3: Spatial Decomposition of Destination Attractiveness

Parameter	Estimate	Interpretation
σ_s^2	0.01687	Spatial persistent variance
ρ	2,929 km	Spatial correlation range (chordal)
σ_a^2	0.00816	Non-spatial persistent variance
σ_ε^2	0.00352	Transitory variance
$\sigma_s^2 + \sigma_a^2 + \sigma_\varepsilon^2$	0.02855	Total variance of u_{it}

6.4 Recovering the Spatial Component

After estimating the variance components, we recover the country-specific spatial and non-spatial persistent components using the Best Linear Unbiased Predictor (BLUP). This approach is closely related to the empirical Bayes and BLUP methods used in the labor economics literature to recover latent persistent heterogeneity from noisy observed outcomes, as in Kane and Staiger (2008) and Chetty et al. (2014). We apply the same basic idea here, using the estimated spatial covariance structure to separate the latent spatial component from non-spatial country-specific heterogeneity and transitory noise. We can express the country mean residuals in vector form,

$$\bar{u} = s + a + \bar{\varepsilon}.$$

Because averaging reduces the variance of the transitory component, the covariance matrix of $\bar{u}$ is

$$V_{\bar{u}} = \Sigma_s + \sigma_a^2 I + D_\varepsilon,$$

where

$$D_\varepsilon = \operatorname{diag}\left(\frac{\sigma_\varepsilon^2}{T_1}, \ldots, \frac{\sigma_\varepsilon^2}{T_N}\right).$$

The spatial component is recovered from its conditional expectation given the observed average residuals:

$$\hat{s} = E[s \mid \bar{u}] = \Sigma_s V_{\bar{u}}^{-1} \bar{u}.$$

Similarly, the non-spatial persistent component is recovered as

$$\hat{a} = E[a \mid \bar{u}] = \sigma_a^2 I V_{\bar{u}}^{-1} \bar{u}.$$

The predictor assigns variation in $\bar{u}$ to the spatial component according to the estimated cross-country covariance structure. Variation that is persistent but not explained by spatial covariance is instead attributed to the non-spatial country component.

As a validation exercise, we construct a leave-one-country-out (LOO) estimate of the spatial component. For each country, we exclude all observations for that country and predict its spatial component using the outcomes of the remaining countries and the estimated spatial covariance structure. The LOO estimates are highly correlated with the full-sample BLUP estimates, with a correlation of 0.944. This close correspondence indicates that the estimated spatial component reflects systematic cross-country spatial dependence rather than simply absorbing country-specific persistent variation.

6.5 Credit Risk and Spatial Dependence in Destination Attractiveness

We investigate whether country credit risk can help explain the spatial dependence in destination attractiveness. Countries with lower credit risk may be more attractive to international investors, holding observed returns fixed. If credit risk is spatially correlated across countries, it may contribute to the spatial structure of destination attractiveness documented above.

Using sovereign credit ratings from Refinitiv Datastream, we find evidence of persistent spatial dependence in country credit risk. Moreover, the estimated cross-spatial covariance between creditworthiness and destination attractiveness is positive in the expanded country sample. These findings suggest that country credit risk may be one economic source of spatial dependence in destination attractiveness. Appendix D provides additional empirical evidence and details the spatial covariance estimation.

7 Quantitative Exercise

For the quantitative exercise, we compute the steady state of the model. To construct steady-state counterparts of the data, we average country-level variables over the 2003–2020 period. The model requires a complete $N \times N$ matrix of bilateral portfolio holdings across investor and destination countries. Because not all countries report bilateral portfolio positions, the quantitative sample is restricted to 52 countries for which this matrix can be constructed. Destinations outside the

quantitative sample are retained through an aggregate Rest-of-the-World outside option.[4]

7.1 Decomposition of Geographic Channels

We first quantify the contribution of the two geographic channels while holding investor-country asset supplies fixed at their baseline levels. Under each counterfactual, portfolio shares adjust, reallocating the fixed global stock of assets across destinations and thereby changing destination capital stocks and equilibrium interest rates. Holding asset supplies fixed isolates the role of geographic forces in reallocating a given global stock of capital, abstracting from the endogenous adjustment in saving implied by the household Euler conditions.

We measure population-weighted interest-rate dispersion relative to the United States as

$$D_{US}(r) = \sum_{i \neq US} w_i \, |r_i - r_{US}|,$$

where w_i denotes country i's population weight among non-U.S. countries. We report the decomposition for the full sample and separately for advanced economies (AE) and emerging market economies (EME).

Figure 3 reports the results. For the full sample, baseline interest-rate dispersion relative to the United States is 1.44 percent, compared with 0.76 percent when both geographic channels are removed. The resulting 0.68 percentage-point contribution is driven mainly by the spatial component of destination attractiveness, while the aggregate contribution of bilateral geographic frictions is close to zero. The remaining increase in dispersion reflects the interaction between the two channels. The aggregate pattern masks substantial heterogeneity across country groups. Among advanced economies, the two geographic channels reduce interest-rate dispersion relative to the United States. Both the bilateral-friction and spatial-attractiveness components contribute negatively to dispersion, although their positive interaction partially offsets these effects. Among emerging market economies, by contrast, geographic forces increase dispersion. This increase is driven primarily by the spatial component of destination attractiveness, with a smaller contribution from bilateral geographic frictions.

[4] Mauritius is included in the Rest-of-the-World outside option because of its small population.

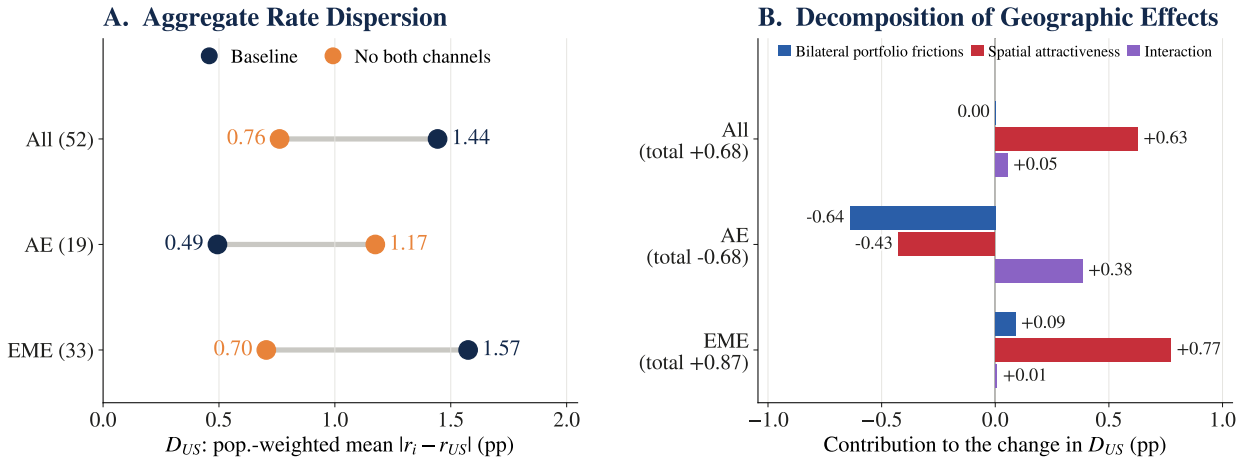


Figure 3: Geographic Channels and Interest-Rate Dispersion Relative to the United States

7.2 Geographic Transmission of Attractiveness Shocks

We quantify the international transmission of the idiosyncratic and spatially correlated attractiveness shocks introduced in Section 4.5, starting from the baseline portfolio allocation. To illustrate the geographic transmission mechanism, we consider a one-percentage-point decline in return-equivalent destination attractiveness originating in Peru. Under an idiosyncratic shock, only Peru's attractiveness declines, whereas a spatially correlated shock also affects other destinations according to the estimated geographic correlation structure.

Figure 4 plots the resulting equilibrium interest-rate responses against geographic distance from Peru. Under the idiosyncratic shock, interest rates decline in all foreign destinations. In contrast, the spatially correlated shock generates positive interest-rate responses in several nearby countries, while more distant countries generally experience negative responses.

7.3 Steady State Counterfactuals

We next solve for the steady-state equilibrium under the same four scenarios, allowing investor-country asset supplies to adjust endogenously through household saving decisions. Unlike the preceding decomposition, which holds asset supplies fixed, this exercise allows both aggregate capital accumulation and its allocation across destinations to respond to changes in the geographic channels. Figure 5 reports the changes in the cross-country variances of log capital and log GDP per capita relative to the baseline.

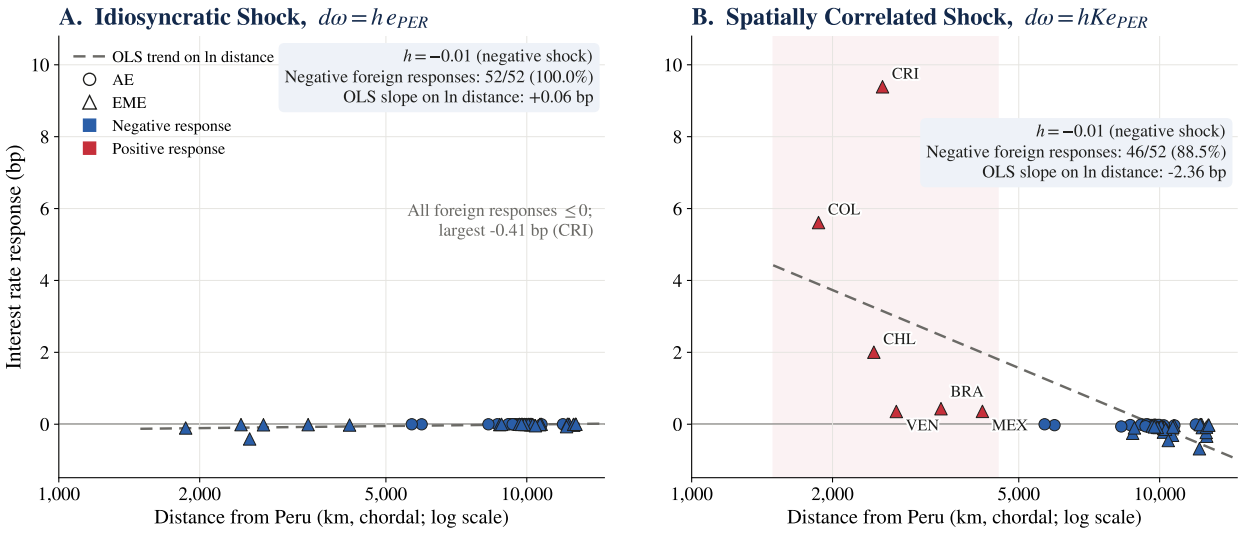


Figure 4: Geographic Transmission of Attractiveness Shocks

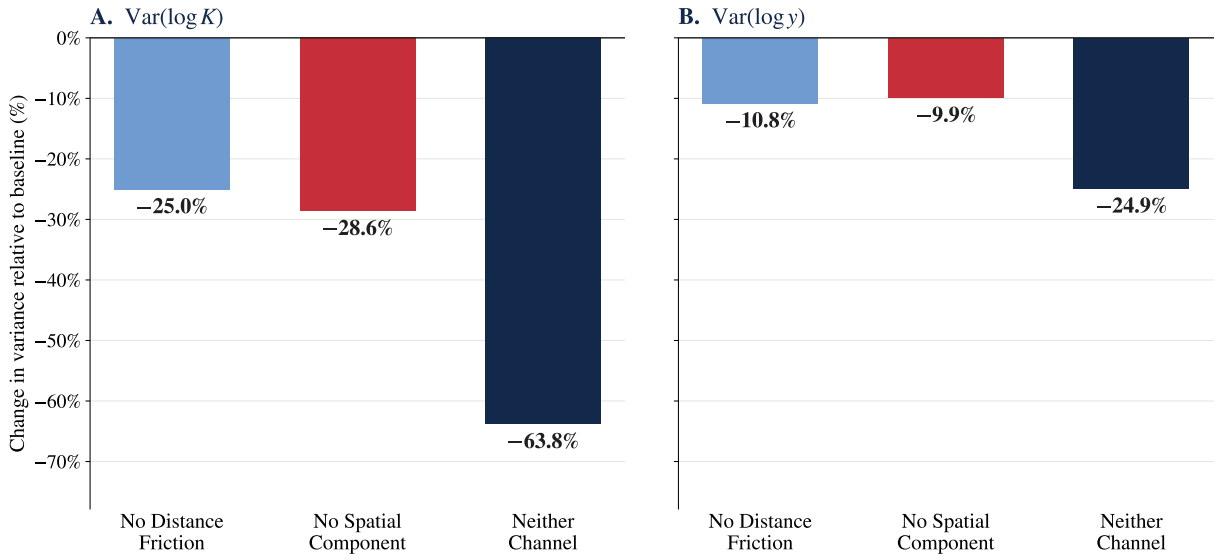


Figure 5: Steady-State Effects of Geographic Channels on Capital and Output Dispersion

Removing bilateral geographic distance frictions reduces the variance of log capital by 25.0 percent and that of log GDP per capita by 10.8 percent. Removing the spatial component of destination attractiveness yields comparable reductions of 28.6 percent and 9.9 percent, respectively.

When both geographic channels are removed, the reductions become substantially larger: 63.8 percent for the variance of log capital and 24.9 percent for the variance of log GDP per capita. These joint effects exceed the sum of the individual effects, indicating that the two channels interact in shaping steady-state outcomes.

Overall, both bilateral portfolio frictions and spatially correlated destination attractiveness contribute substantially to cross-country differences in capital allocation and output.

8 Conclusion

This paper investigates how geography shapes international capital allocation and interest rate differences. We document that geographically distant countries exhibit larger interest rate differences, including in their persistent component. To explain this pattern, we develop a multicountry general equilibrium model that distinguishes bilateral portfolio frictions from spatially correlated destination attractiveness.

Our quantitative analysis reveals distinct roles for these two geographic channels. Spatially correlated destination attractiveness accounts for most of the geographic contribution to aggregate interest rate dispersion, with contrasting effects across advanced and emerging economies. The spatial structure of attractiveness shocks can also reverse the direction of international interest rate transmission, while bilateral portfolio frictions primarily affect the magnitude of these responses. In steady state, both geographic channels contribute substantially to the dispersion of capital and output across countries. These findings highlight that geography matters for international finance not only through bilateral barriers to investment but also through spatial dependence across investment destinations.

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Appendix

A Additional Empirical Results

A.1 Bilateral Portfolio Friction Estimation

Table 4 reports the PPML estimates of bilateral portfolio frictions, including geographic distance, home bias, and geopolitical alignment.

Table 4: Bilateral Portfolio Frictions and Geographic Distance

	(1)
	share
$\ln(dist_{i,j})$	-0.797***
	(0.0544)
Foreign	-4.159***
	(0.130)
Cross Block	-0.796***
	(0.155)
Observations	100377

Standard errors in parentheses

Standard errors clustered by country-pair (country x counterpart).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.2 σ_N Estimation

The return term is endogenous because equilibrium interest rates are jointly determined with international portfolio allocation. To estimate σ_N, we use an IV strategy that builds on Koijen and Yogo (2026) and adapts their residual-supply approach to our international capital-allocation setting. The instrument exploits variation in predicted asset demand relative to predicted asset supply. This residual demand pressure shifts equilibrium returns through asset-market clearing but, under the exclusion restriction, does not directly affect investor n's bilateral portfolio demand. It therefore provides exogenous variation in returns that identifies the sensitivity of portfolio allocation to equilibrium returns.

Equation (3) can be modified to be expressed as

$$\log \frac{b_{ni}}{b_{nn}} = \frac{1}{\sigma_N}(r_{it} - r_{nt} - \Phi_{ni} + \omega_{it} - \omega_{nt})$$

Let X_{ni} denote the observed bilateral characteristics between investor country n and destination country i, including bilateral geographic distance and the cross-bloc indicator. The identification assumptions for the IV strategy are:

$$\begin{aligned} \text{Exogeneity:} \quad & \operatorname{Cov}\left(\varepsilon_{nit},\, dist_{ji} \mid X_{ni}, \iota_i, \eta_n\right) = 0, \\ \text{Relevance:} \quad & \operatorname{Cov}\left(r_{it},\, dist_{ji} \mid X_{ni}, \iota_i, \eta_n\right) \neq 0, \qquad j \neq n. \end{aligned}$$

The exogeneity condition requires that, conditional on the bilateral characteristics between investor country n and destination country i, the geographic distance between other investor countries j and destination i is uncorrelated with the idiosyncratic portfolio-demand shock of investor n. Intuitively, the location of other investor countries should not directly affect country n's relative preference for destination i.

The relevance condition requires that the geographic distribution of other investor countries affects the equilibrium return in destination i. Because bilateral portfolio demand declines with geographic distance, destinations that are relatively close to countries with greater investment capacity face stronger predicted demand. Through capital-market clearing, this variation in demand pressure affects the equilibrium return r_{it}. We aggregate this cross-country distance variation into the residual-demand instrument constructed below.

Demand Component From equation (3), we construct the time-invariant component of portfolio demand using only the estimated bilateral gravity frictions. Specifically, the predicted allocation share from investor country m to destination country i is given by

$$\hat{b}_{mi}^{gravity} = \frac{\exp(-\phi_{mi}^{gravity})}{\sum_h \exp(-\phi_{mh}^{gravity})},$$

where

$$\phi_{ni}^{gravity} = \hat{\beta}^{dist} \ln(dist_{ni}) + \hat{\beta}^{foreign} \mathbb{1}[n \neq i].$$

This predicted share captures the time-invariant geographic structure of portfolio demand implied by bilateral distance and home bias, abstracting from time-varying destination characteristics and

equilibrium returns. We then aggregate predicted demand from all investor countries other than n:

$$\sum_{m \neq n} A_{mt} \hat{b}_{mi}^{gravity}.$$

Supply Component To construct the supply component, we estimate the following panel regression:

$$\log S_{it} = \Psi' z_{it} + \lambda_t + \varepsilon_{it}^S,$$

where $z_{it} = [\log GDP_{it}, \log pop_{it}]'$, and λ_t denotes year fixed effects. We then construct predicted asset supply as

$$\log \hat{S}_{it} = \hat{\Psi}' z_{it} + \hat{\lambda}_t.$$

The predicted value captures the component of asset supply explained by observable country characteristics and common time variation.

Residual Demand Pressure Using the predicted asset allocation from the gravity model, we first construct a leave-one-out measure of demand pressure for each destination i:

$$Z_{i,-n,t} = \log \left(\sum_{m \neq n} A_{mt} \hat{b}_{mit}^{gravity} \right) - \log \hat{S}_{it}.$$

This measure is the negative of the conventional residual-supply measure. A larger value of $Z_{i,-n,t}$ therefore indicates stronger predicted demand relative to asset supply in destination country i. The leave-one-out construction excludes predicted demand originating from investor country n, so that the bilateral portfolio choice of investor n does not mechanically enter the demand component. Because the endogenous regressor is the return differential between destination i and country n, we use the corresponding difference in residual demand pressure as the instrument:

$$Z_{nit}^{rel} = Z_{i,-n,t} - Z_{n,-n,t}.$$

Thus, Z_{nit}^{rel} measures the relative predicted demand pressure on destination i compared with country n, generated by investors other than n. Variation in this relative demand pressure shifts the equilibrium return differential $r_{it} - r_{nt}$ through asset-market clearing.

2SLS The return elasticity of portfolio demand is estimated using two-stage least squares. Since the portfolio-demand equation is expressed in terms of returns relative to the investor's domestic asset, we instrument the return differential $r_{it} - r_{nt}$ with the corresponding difference in leave-one-out residual demand pressure. The first stage is

$$r_{it} - r_{nt} = \pi Z^{rel}_{nit} + \mathbf{X}'_{ni}\beta + \nu_{nit},$$

where

$$Z^{rel}_{nit} = Z_{i,-n,t} - Z_{n,-n,t}.$$

The estimated return coefficient implies $\hat{\sigma}_N = \frac{1}{\hat{\lambda}_r} \approx 0.075$

Table 5: IV Estimation Results

	(1)	(2)
	First stage	2SLS
Relative Z	0.01***	
	(0.00)	
ΔR		13.58***
		(3.39)
First-stage F (excl. instr.)		688.33
Kleibergen-Paap rk Wald F		688.33
Observations	28772	28772
Adjusted R^2		0.092

Fixed effects: country and counterpart. Robust standard errors in parentheses.

B Robustness Checks

We examine whether the estimated effects of geographic distance and geopolitical alignment on portfolio allocation are robust to alternative specifications. Recall that the bilateral wedge ϕ_{ni} enters portfolio demand with a negative sign. Thus, a negative coefficient in the portfolio allocation equation corresponds to a larger portfolio wedge.

First, we augment the baseline specification with additional bilateral covariates, including exchange-rate volatility, common language, and trade intensity. Table 6 reports the baseline estimate in the

full sample, the baseline estimate in the common sample, and specifications that sequentially add these controls. The coefficients on log geographic distance and the cross-block indicator remain negative and statistically significant across specifications.

Table 6: Robustness of Bilateral Portfolio Friction Estimates

	(1)	(2)	(3)	(4)	(5)	(6)
$\ln(dist_i,j)$	-0.797***	-0.790***	-0.716***	-0.726***	-0.706***	-0.650***
	(0.054)	(0.055)	(0.052)	(0.058)	(0.063)	(0.062)
Foreign	-4.159***	-4.171***	-3.730***	-4.424***	-4.478***	-3.991***
	(0.130)	(0.131)	(0.117)	(0.144)	(0.156)	(0.148)
Cross Block	-0.796***	-0.803***	-0.705***	-0.773***	-0.819***	-0.722***
	(0.155)	(0.157)	(0.172)	(0.156)	(0.157)	(0.171)
FX volatility (%)			-0.288***			-0.279***
			(0.030)			(0.030)
Common language				0.661***		
				(0.169)		
Trade intensity (%)					0.141***	0.108***
					(0.039)	(0.036)
Sample	Full	Common	Common	Common	Common	Common
Observations	100377	95086	95086	95086	95086	95086

Standard errors in parentheses

Origin×year and destination×year fixed effects. Standard errors clustered by country pair.

FX volatility: s.d. of monthly bilateral exchange-rate returns within the year (%).

Trade intensity: (X_ij+X_ji)/(GDP_i+GDP_j) (%), IMF IMTS and World Bank.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

We also allow home bias to vary across investor countries. Specifically, we parameterize the bilateral wedge as

$$\phi_{ni} = \beta^{dist} \ln(dist_{ni}) + \beta_n^{foreign} \mathbb{1}[n \neq i] + \beta^{cross\text{-}block} CrossBlock_{ni}.$$

Allowing for investor-specific home bias leaves the main results unchanged. The coefficients on log geographic distance and the cross-block indicator in the portfolio allocation equation remain negative and statistically significant.

Table 7: Robustness to Investor-Specific Home Bias

	(1) share
Log distance	-0.921***
	(0.045)
Cross block	-0.579***
	(0.216)
Observations	100377.000

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

C Validation of Spatial Attractiveness

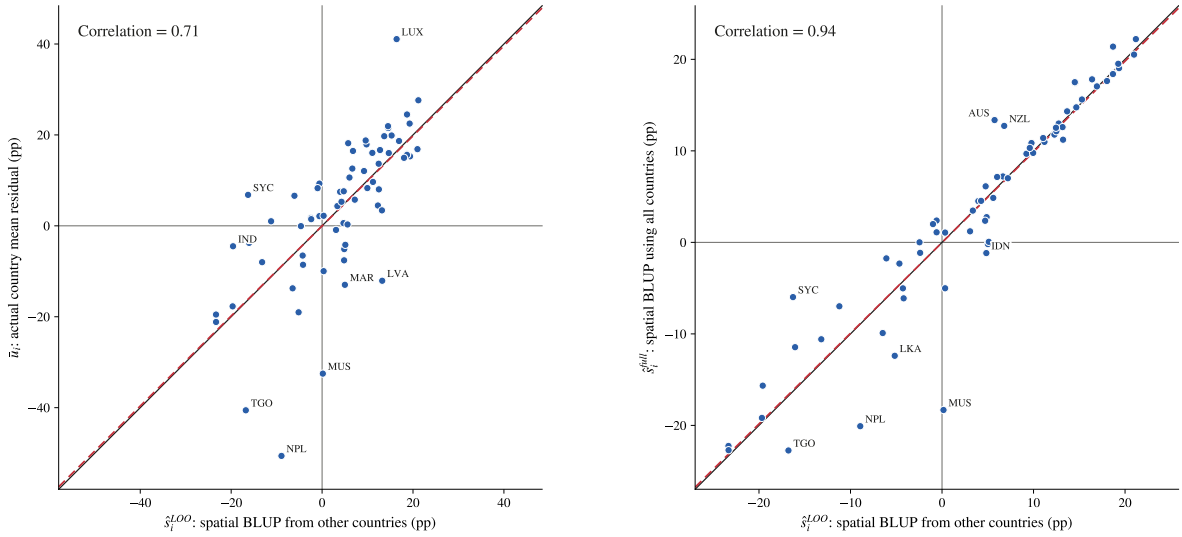
We assess the estimated spatial component of destination attractiveness using a leave-one-country-out (LOO) procedure. For each country, we construct a spatial BLUP using information from all other countries, excluding its own residual from the prediction.

Figure 6 presents two validation exercises. Panel A compares the LOO spatial predictions with observed country-level mean residuals. The two measures are positively correlated (0.71), suggesting that the geographic structure of destination attractiveness allows country-level differences to be predicted using information from other countries.

Panel B compares the full-sample and LOO spatial BLUPs. The two estimates are highly correlated (0.94), indicating that the estimated spatial component is not primarily driven by each country's own residual but instead reflects geographic information shared across countries.

D Credit Risk and Spatial Dependence

This appendix provides additional evidence on the geographic structure of sovereign credit risk and its relationship with destination attractiveness. We examine whether sovereign credit risk exhibits persistent spatial dependence and whether its spatial component is associated with the spatial structure of destination attractiveness documented in the main text. Throughout the analysis, we measure country credit risk using sovereign credit-rating scores from Refinitiv Datastream, where higher values indicate greater creditworthiness and therefore lower credit risk.



(a) Prediction from Other Countries

(b) Full-Sample vs. LOO Spatial BLUPs

Figure 6: Validation of Spatial Attractiveness

D.1 Geographic Patterns in Sovereign Credit Risk

We first examine whether bilateral differences in sovereign credit risk vary systematically with geographic distance. We consider two measures of sovereign credit risk, sovereign credit-rating scores and CDS spreads. For each country, we compute the average credit-rating score and average log CDS spread over 2012–2023. Following the empirical specification for interest-rate differentials and destination attractiveness, we estimate

$$|\bar{Z}_i^k - \bar{Z}_j^k| = \beta^k \ln(dist_{ij}) + \alpha_i^k + \alpha_j^k + \varepsilon_{ij}^k, \tag{15}$$

where $k \in \{C, CDS\}$ denotes the measure of sovereign credit risk. $\bar{Z}_i^C$ represents the average sovereign credit-rating score of country i, while $\bar{Z}_i^{CDS}$ denotes its average log CDS spread.

Table 8 reports the estimation results. Geographic distance is positively and significantly associated with bilateral differences in both sovereign credit-rating scores and log CDS spreads. These findings indicate that geographically distant countries tend to exhibit larger differences in sovereign credit risk, consistent with the geographic patterns documented for destination attractiveness in the main text.

Table 8: Geographic Distance and Sovereign Credit Risk Differences

	(1) Sov. score gap	(2) CDS gap
Log distance	0.317***	0.191***
	(0.030)	(0.024)
Origin FE	Yes	Yes
Destination FE	Yes	Yes
Countries	145	67
Observations	10,438	2,209
R^2	0.299	0.529

Robust standard errors in parentheses.
* p<0.10, ** p<0.05, *** p<0.01

D.2 Sovereign Credit Risk and Spatial Destination Attractiveness

The geographic patterns in sovereign credit risk documented above suggest that creditworthiness may be one economic characteristic underlying the spatial dependence in destination attractiveness. To examine this possibility, we jointly model the persistent components of sovereign creditworthiness and destination attractiveness. We focus on sovereign credit-rating scores, which provide broader country coverage than CDS spreads in the joint estimation sample. The rating score is increasing in sovereign creditworthiness. The joint estimation uses the overlapping 2012–2020 period.

D.2.1 Joint Spatial Covariance

Following the spatial decomposition of destination attractiveness in the main text, we specify the bivariate model

$$\begin{pmatrix} C_{it} \\ \omega_{it} \end{pmatrix} = \begin{pmatrix} \alpha_t^C \\ \alpha_t^\omega + \gamma_P \ln Pop_{it} \end{pmatrix} + \begin{pmatrix} s_i^C \\ s_i^\omega \end{pmatrix} + \begin{pmatrix} a_i^C \\ a_i^\omega \end{pmatrix} + \begin{pmatrix} \varepsilon_{it}^C \\ \varepsilon_{it}^\omega \end{pmatrix}, \tag{16}$$

where C_{it} denotes the sovereign credit-rating score, s_i^C and s_i^ω are persistent spatial components, and a_i^C and a_i^ω capture persistent non-spatial country heterogeneity. We allow the non-spatial country components to be correlated across the two outcomes.

We model the joint spatial covariance as

$$\operatorname{Cov}\left(\begin{pmatrix} s_i^C \\ s_i^\omega \end{pmatrix}, \begin{pmatrix} s_j^C \\ s_j^\omega \end{pmatrix}\right) = \exp\left[-\left(\frac{d_{ij}}{\rho}\right)^2\right] \underbrace{\begin{pmatrix} \sigma_{s,C}^2 & \sigma_{s,C\omega} \\ \sigma_{s,C\omega} & \sigma_{s,\omega}^2 \end{pmatrix}}_{\Sigma_s}, \tag{17}$$

where ρ is the common geographic correlation range. The parameter $\sigma_{s,C\omega}$ measures the spatial cross-covariance between sovereign creditworthiness and destination attractiveness. The corresponding spatial correlation is

$$r_s = \frac{\sigma_{s,C\omega}}{\sqrt{\sigma_{s,C}^2 \sigma_{s,\omega}^2}}. \tag{18}$$

We estimate the joint spatial covariance structure using restricted maximum likelihood. Table 9 reports the results. The estimates indicate a positive spatial relationship between sovereign creditworthiness and destination attractiveness. This suggests that geographically correlated creditworthiness may be one economic characteristic underlying the persistent spatial dependence in destination attractiveness.

Table 9: Joint Spatial Covariance of Sovereign Creditworthiness and Destination Attractiveness

Sample	Countries	$\hat{\sigma}_{s,C\omega}$	$\hat{r}_s$	LR	Bootstrap p
Baseline	65	0.238	1.000	5.21	0.025
Excluding Venezuela	64	0.242	1.000	13.99	0.005
At least 5 observations	59	0.251	0.999	8.38	0.005

Notes: The table reports joint spatial REML estimates for sovereign creditworthiness and destination attractiveness over 2012–2020. The likelihood-ratio test examines the null of zero spatial cross-covariance. Bootstrap p-values are based on 199 simulations. The minimum-observation restriction refers to destination attractiveness.

E Geographic and Geopolitical Dependence

Table 2b provides evidence that countries' estimated latent attractiveness is associated with both geographic and geopolitical proximity, with the latter measured by distance in UN General Assembly voting ideal points. To capture these two sources of dependence, we specify an additive Gaussian covariance kernel:

$$\operatorname{Cov}(s_i, s_j) = \sigma_G^2 \exp\left[-\left(\frac{\tilde{d}_{G,ij}}{\rho_G}\right)^2\right] + \sigma_P^2 \exp\left[-\left(\frac{\tilde{d}_{P,ij}}{\rho_P}\right)^2\right]. \tag{19}$$

Here, the subscripts G and P denote the geographic and geopolitical components, respectively. Geographic distance is measured by chordal distance, while geopolitical distance is the absolute difference between countries' sample-period average ideal points. Each distance is divided by its standard deviation across distinct unordered country pairs, without subtracting its mean. The parameters σ_G^2 and σ_P^2 govern the component variances, while ρ_G and ρ_P govern their correlation ranges.

This specification corresponds to the decomposition $s_i = s_i^G + s_i^P$, with $\mathrm{Cov}(s_i^G, s_j^P) = 0$ for all i, j. This restriction concerns the latent components and does not require geographic and geopolitical distances to be uncorrelated. We also retain the independent country-specific component a_i, with variance σ_a^2.

Using the geopolitical specification of ω_{it}, with 1,028 country-year observations from 67 countries and controlling for log population and year fixed effects, we estimate $\hat{\sigma}_G^2 = 0.00552$, $\hat{\sigma}_P^2 = 0.01501$, and $\hat{\sigma}_a^2 = 0.00791$. These components account for approximately 19.4%, 52.8%, and 27.8% of the model-implied persistent variance, respectively. The estimated decomposition therefore attributes a substantial share of persistent heterogeneity to geopolitical dependence, while retaining geographic dependence and an independent country-specific component.

Converted to the original distance units, the estimated geographic and geopolitical ranges are 1,677 km and 1.366 ideal-point units, respectively. At these distances, the correlation within the corresponding component falls to e^{-1}, approximately 0.37. The geographic range is shorter than the 2,929 km estimated under the geography-only covariance specification, indicating a reallocation of covariance when geopolitical dependence is modeled separately.

A parametric bootstrap test with 499 replications rejects the null hypothesis $H_0 : \sigma_P^2 = 0$ at the 5% level ($p = 0.012$). This result provides evidence of geopolitical dependence beyond that captured by the geographic covariance component.

F Destination Attractiveness and Country Characteristics

Table 10 examines how the recovered destination attractiveness, $\omega_{i,t}$, is related to observable country characteristics. By construction, $\omega_{i,t}$ captures components of destination attractiveness other than interest rates.

Population is positively associated with destination attractiveness across countries, consistent with larger markets attracting greater portfolio demand. Sovereign creditworthiness is also positively and significantly associated with destination attractiveness across specifications, including those with country fixed effects. This suggests that countries with stronger sovereign credit ratings tend to attract greater portfolio demand conditional on observed returns.

Table 10: Destination Attractiveness and Country Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)
ln(Population)	0.055***		0.059***	-0.439**		-0.406*
	(0.012)		(0.011)	(0.204)		(0.206)
Sovereign Credit Score		0.030***	0.034***		0.011**	0.008*
		(0.008)	(0.006)		(0.005)	(0.004)
Country FE	No	No	No	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	531	531	531	528	528	528
Countries	66	66	66	63	63	63
Adj. R^2	0.281	0.132	0.473	0.960	0.958	0.960

Standard errors clustered by country in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

G Model Solution Algorithm

G.1 Solution Algorithm

We solve the stationary general equilibrium using a nested numerical procedure. All parameters, including the calibrated discount factors $\{\beta_n\}$, are held fixed throughout the solution. We use the normalized bilateral portfolio wedge

$$\phi_{ni} \equiv \frac{\Phi_{ni}}{\sigma_N},$$

as defined in Section 5.1.

For a given vector of asset supplies $\{A_n\}$, the inner loop solves for capital allocation, portfolio shares, and interest rates. Given a current guess for capital $\{k_i\}$, interest rates are

$$r_i = \alpha Z_i \left(\frac{k_i}{L_i}\right)^{\alpha-1} \tag{20}$$

Portfolio shares are then determined by

$$b_{ni} = \frac{\exp\left(-\phi_{ni} + \omega_i/\sigma_N + r_i/\sigma_N\right)}{\sum_j \exp\left(-\phi_{nj} + \omega_j/\sigma_N + r_j/\sigma_N\right)}, \tag{21}$$

with the modification for destinations outside the model sample described in Appendix G.2. The implied capital stock is

$$k_i' = \sum_n b_{ni} A_n. \tag{22}$$

We iterate over $\{k_i\}$ until the implied capital allocation converges.

Given the resulting portfolio allocation and interest rates, the outer loop adjusts asset supplies until the steady-state Euler condition

$$\beta_n R_n = 1 \tag{23}$$

is satisfied for every investor country, where

$$R_n = 1 + \sum_i b_{ni} r_i. \tag{24}$$

Algorithm 1 Solution of the Stationary General Equilibrium

Initialize asset supplies $\{A_n\}$.

repeat

  Initialize destination capital stocks $\{k_i\}$.

  repeat

    Compute interest rates:

$$r_i = \alpha Z_i \left(\frac{k_i}{L_i}\right)^{\alpha-1}$$

    Compute bilateral portfolio shares $\{b_{ni}\}$.

    Update destination capital:

$$k_i' = \sum_n b_{ni} A_n.$$

  until $\{k_i\}$ converges

  Compute investor portfolio returns:

$$R_n = 1 + \sum_i b_{ni} r_i.$$

  Update $\{A_n\}$ to satisfy

$$\beta_n R_n = 1 \qquad \forall n.$$

until the Euler conditions converge

return $\{A_n, b_{ni}, k_i, r_i\}$.

G.2 Treatment of the Rest of the World

The set of destinations observed in the bilateral portfolio data is larger than the set of countries explicitly included in the general-equilibrium model. Dropping these destinations would mechanically reallocate their portfolio shares toward countries in the model sample. We therefore aggregate all destinations outside the model sample into a single outside option.

Let $\mathcal{N}$ denote the set of model countries and $\mathcal{R}$ the set of observed destinations outside $\mathcal{N}$. For

investor n, we construct the inclusive value

$$IV_{nR} = \log \sum_{j \in \mathcal{R}} \exp\left(-\phi_{nj} + v_j\right), \tag{25}$$

where v_j is the estimated destination attractiveness component. The portfolio problem then treats IV_{nR} as an additional investment alternative:

$$b_{ni} = \frac{\exp(V_{ni})}{\sum_{j \in \mathcal{N}} \exp(V_{nj}) + \exp(IV_{nR})}, \qquad V_{ni} = -\phi_{ni} + \frac{\omega_i + r_i}{\sigma_N}. \tag{26}$$

Assets allocated to the outside option do not enter the capital stock of countries in $\mathcal{N}$. Hence, capital-market clearing within the model is

$$k_i = \sum_{n \in \mathcal{N}} b_{ni} A_n, \qquad i \in \mathcal{N}. \tag{27}$$

Because the model does not solve for an endogenous return outside the model sample, portfolio returns entering the Euler equation are computed using returns on model destinations only.

H Additional Quantitative Results

H.1 Model Fit

Figure 7 compares the model-implied interest rates with their empirical counterparts. The model reproduces the overall cross-country variation in interest rates.

H.2 Spatial Decomposition of Destination Attractiveness

Figure 8 presents the estimated persistent components of destination attractiveness. Panel (a) shows the spatial component, $\hat{s}_i$, while Panel (b) shows the non-spatial country-specific component, $\hat{a}_i$. Both components are recovered using the full-sample BLUP estimates described in Section 6.4.

H.3 Counterfactual Reallocation with Fixed Asset Stocks

Figure 9 shows how capital is reallocated across destinations when each geographic channel is removed while country-level asset stocks are held fixed at their baseline values. Conditional on these asset stocks, portfolio shares, destination capital stocks, and equilibrium interest rates are jointly re-solved.

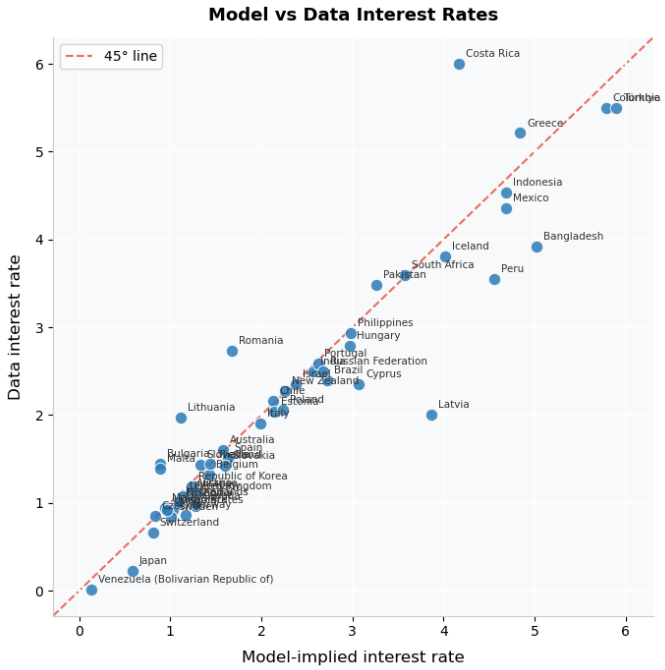
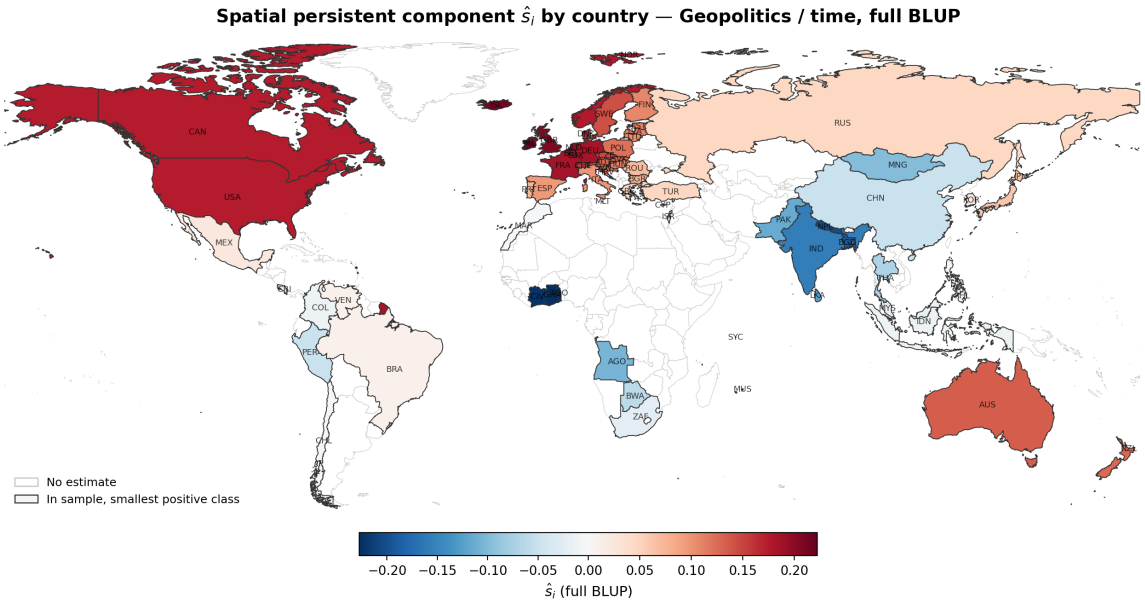


Figure 7: Model-Implied and Observed Interest Rates

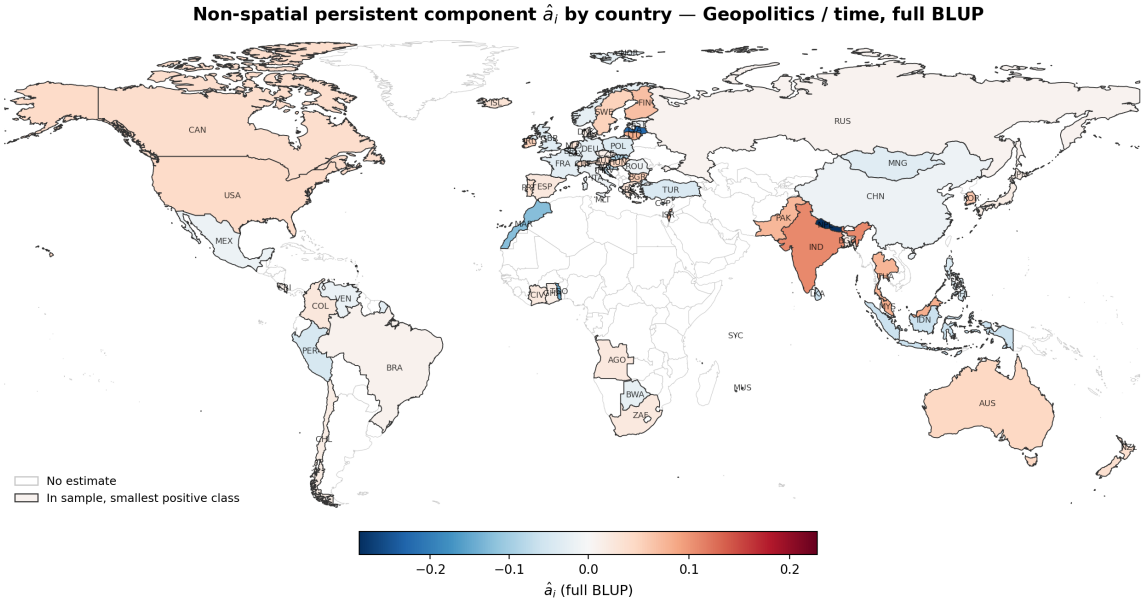
Figure 10 reports the corresponding changes in interest rates. Each point compares the baseline interest rate with the counterfactual interest rate for the same country. Points above the 45-degree line indicate an increase in the equilibrium interest rate relative to the baseline, while points below the line indicate a decrease.

H.4 Geographic Transmission of Attractiveness Shocks

Figure 11 illustrates how interest-rate responses vary with geographic distance between the shock origin and recipient countries.



(a) Spatial persistent component, $\hat{s}_i$



(b) Non-spatial persistent component, $\hat{a}_i$

Figure 8: Decomposition of Destination Attractiveness

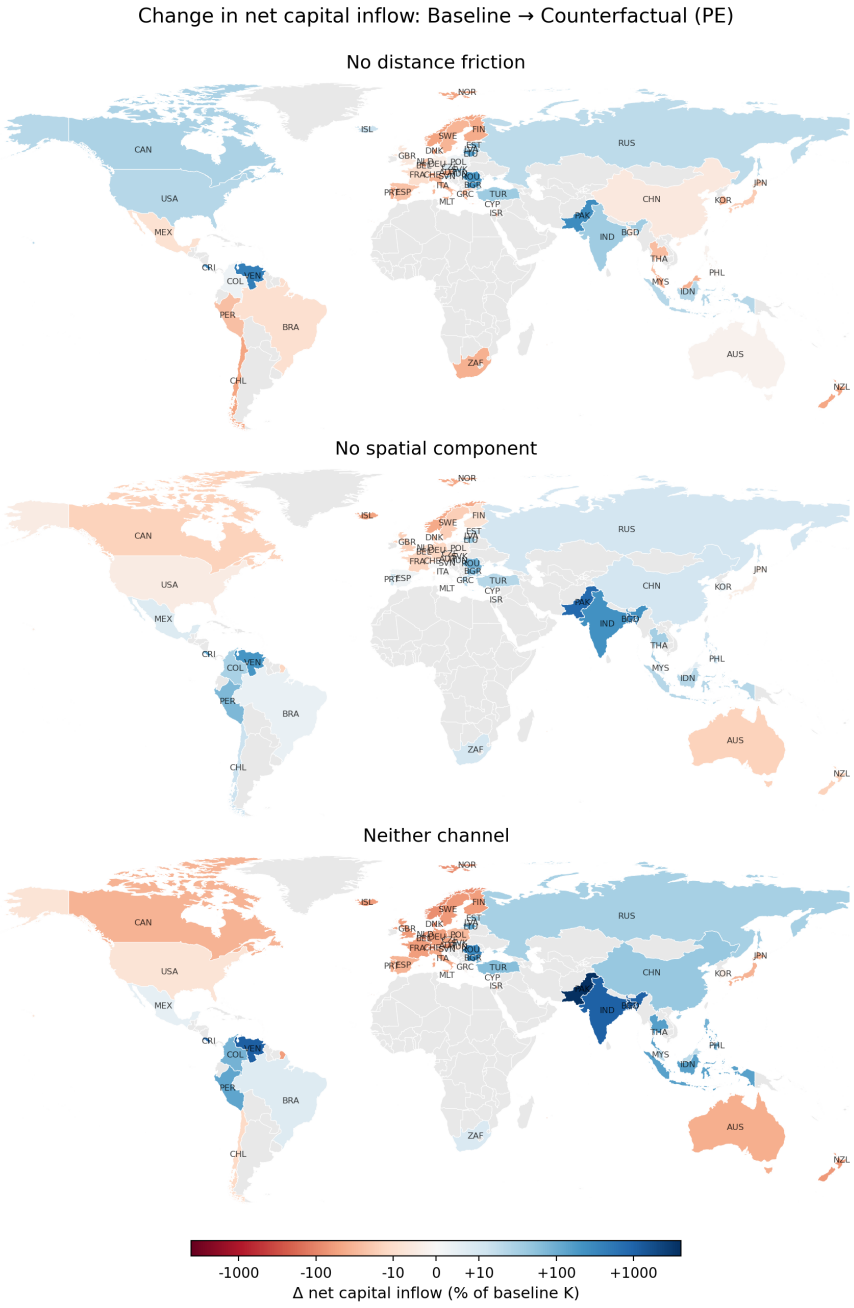


Figure 9: Capital Reallocation with Fixed Asset Stocks

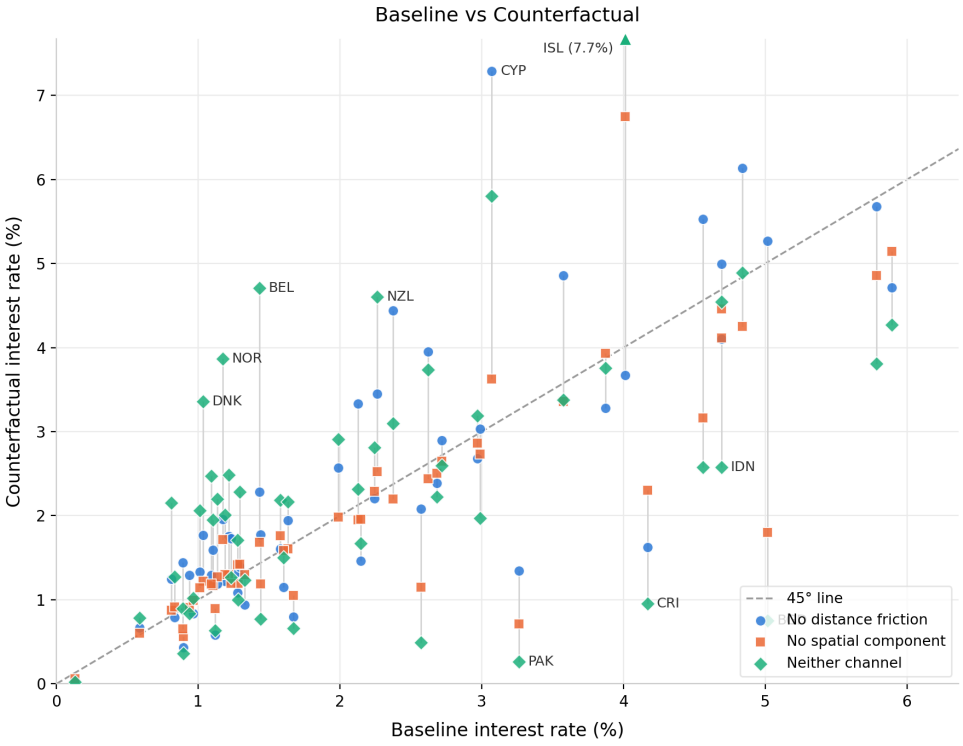


Figure 10: Interest Rate Responses with Fixed Asset Stocks

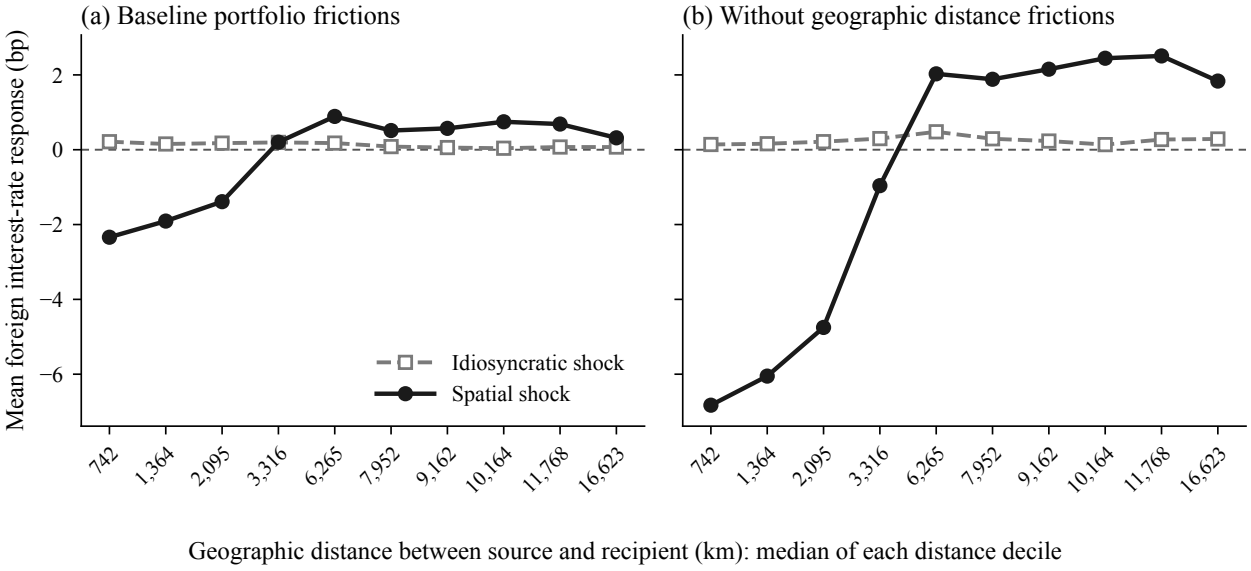


Figure 11: Interest-Rate Responses by Geographic Distance

Notes: The figure plots mean foreign interest-rate responses across deciles of bilateral geographic distance. We consider each of the 53 countries as a shock origin and compute responses for all 2,756 directed source-recipient country pairs. Shocks are normalized to a one-percentage-point increase in return-equivalent attractiveness in the origin country. Responses are based on local general-equilibrium comparative statics and expressed in basis points. Panel (a) reports results under baseline portfolio frictions, while Panel (b) removes the geographic distance component of bilateral portfolio frictions.

I Proofs

I.1 Proof of Proposition 1

Proof. Consider an interior equilibrium with fixed investor asset stocks $\{A_n\}$, strictly positive portfolio shares, and decreasing marginal products of capital.

Equilibrium Interest-Rate Transmission. Let $\boldsymbol{k}$ denote the vector of destination capital stocks. The equilibrium capital-market clearing condition is

$$\boldsymbol{k} = \boldsymbol{S}(\boldsymbol{r} + \boldsymbol{\omega}), \tag{28}$$

where $\boldsymbol{S}(\cdot)$ denotes aggregate capital demand across investment destinations implied by portfolio choices.

Totally differentiating the market-clearing condition gives

$$d\boldsymbol{k} = J(d\boldsymbol{r} + d\boldsymbol{\omega}), \tag{29}$$

where

$$J = \frac{\partial \boldsymbol{S}}{\partial(\boldsymbol{r} + \boldsymbol{\omega})} \tag{30}$$

is the Jacobian of aggregate portfolio demand.

From the firms' optimality conditions, equilibrium interest rates are decreasing in destination capital stocks:

$$d\boldsymbol{r} = -H\, d\boldsymbol{k}, \tag{31}$$

where

$$H = \operatorname{diag}\left(-\frac{\partial r_i}{\partial k_i}\right)_{i=1}^{N} \tag{32}$$

is a positive diagonal matrix.

Substituting the capital-market clearing condition yields

$$d\boldsymbol{r} = -HJ(d\boldsymbol{r} + d\boldsymbol{\omega}). \tag{33}$$

Rearranging,

$$(H^{-1} + J)d\boldsymbol{r} = -J\, d\boldsymbol{\omega}. \tag{34}$$

Therefore,

$$d\boldsymbol{r} = -P\, d\boldsymbol{\omega}, \qquad P = (H^{-1} + J)^{-1} J. \tag{35}$$

This establishes the general equilibrium interest-rate transmission relationship.

Properties of the Transmission Matrix. Under the logit portfolio allocation rule, aggregate capital demand for destination i is

$$S_i = \sum_n A_n b_{ni}. \tag{36}$$

The elements of the Jacobian J are

$$J_{ij} = \frac{1}{\sigma_N} \sum_n A_n b_{ni} \left(\mathbb{1}[i = j] - b_{nj} \right). \tag{37}$$

Since portfolio shares are strictly positive,

$$J_{ii} > 0, \qquad J_{ij} < 0 \quad (i \neq j). \tag{38}$$

Moreover, because portfolio shares sum to one,

$$J\mathbf{1} = 0. \tag{39}$$

Define

$$M = H^{-1} + J. \tag{40}$$

Since H^{-1} is positive diagonal and J is positive semidefinite with nonpositive off-diagonal elements, M is a symmetric positive definite Stieltjes matrix.

Consequently, M^{-1} is elementwise nonnegative and, under strictly positive portfolio shares, strictly positive.

Using

$$P = (H^{-1} + J)^{-1} J = I - M^{-1} H^{-1}, \tag{41}$$

we obtain

$$P_{ij} = -(M^{-1})_{ij} (H^{-1})_{jj} < 0, \qquad i \neq j. \tag{42}$$

Furthermore, since $J\mathbf{1} = 0$,

$$P\mathbf{1} = 0. \tag{43}$$

It follows that

$$P_{ii} = -\sum_{j \neq i} P_{ij} > 0. \tag{44}$$

Since

$$P_{ii} = 1 - (M^{-1})_{ii}(H^{-1})_{ii}, \tag{45}$$

we also have $P_{ii} < 1$.

Therefore,

$$0 < P_{ii} < 1, \qquad P_{ij} < 0 \quad (i \neq j). \tag{46}$$

This completes the proof.

□